

Video 17 – Three Learning Principles

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Video Synopsis

Introduction **04:55 – 06:23**

Prof Abu-Mostafa starts by indicating that, after quite a maths heavy set of lectures, this lecture will be more philosophical focusing on three learning principles: Occam's Razor, Sampling (Data) bias and Data Snooping.

Occam's Razor **06:23 – 23:00**

Prof. Abu-Mostafa introduces complexity in two ways: 1) as that of an object, in our case one hypothesis, and 2) the complexity of a family of objects, in our case a family of hypotheses. It is demonstrated that these can be quite different. Applied to machine learning: the VC dimension is a measure of complexity of a family of hypotheses and can be very high, even if the complexity of individual hypothesis in that family is much lower. When we think of Occam's Razor to say 'simple is better' we think of the complexity of an object / hypothesis.

To illustrate the difference, Prof Abu-Mostafa introduces a postal scam in which people are sent 5 letters with predictions (win/lose) of future football results. As the scammer sends letters with both outcomes, and always only continues sending letters to those who received the correct outcome, there will be a person (assuming 32 letters were sent originally) who will have received all correct predictions. Hence, from this person's perspective it may seem like a great deal to pay \$50 for a final letter, in which the scammer promises to provide the outcome of a 6th match (or not send the letter at all of course; after all this was a scam). This example shows that what may seem like a very simple hypothesis set (the hypothesis being that the scammer can really see the future) is often actually a much more complex hypothesis set. This is important to realise, because it also shows that, if a hypothesis set is truly simple, and it fits the data, it is more likely to be truly a good fit than a large hypothesis set fitting the data, as this could be by chance (or a scam as in the example).

SAFE SKIP: Examples of meaningless fits **28:30 – 30:43**

In a quick example Prof. Abu-Mostafa once more shows that data is king by showing that you cannot demonstrate that a process is linear if you have only two data points. Such an experiment results in a lack of falsifiability, or the ability to show that something cannot be correct. This is very similar to our requirement for sufficient data to ensure models do not overfit and generalise well.

Sampling (Data) Bias **30:43 – 45:05**

Prof. Abu-Mostafa introduces data bias using an example from the US elections. In 1948 the Chicago Daily Tribune declared Dewey the victor in the US presidential elections over Truman. However, due to the sample being gathered using phone interviews, the paper biased its sample towards those with a phone. In 1948 this meant sampling a more affluent population than average which leaned towards Dewey. The resulting lesson is that:

If the data is sampled in a biased way, learning will produce a similarly biased outcome.

Prof. Abu-Mostafa introduces matching of the train and test distributions as a solution, but limited by the fact that you often do not know the distribution of the test data (where this is real out-of-sample data). Where you do know characteristics of the test data, you can try to resample the training data to match the distribution of the test data, but this is beyond the scope of this module. The important lesson in this section is that the whole mathematical machinery around the VC dimension assumed that the probabilistic characteristics of train and test data were the same. Where this assumption is violated, the VC theory does no longer hold.

This section concludes with a further example (Puzzle 3) of data bias in a bank loan scenario. Data bias is introduced here because data sets that contain information on the profitability of historic loans, will per definition not contain those individuals who were refused for a loan.

Data Snooping

45:05 – 1:00:08

Data snooping is introduced as tricky as it can be introduced very easily and without clear evidence of snooping. The danger of snooping is repeated from previous lectures as:

If a data set has affected any step in the learning process, its ability to assess the outcome has been compromised.

Prof. Abu-Mostafa states that data snooping is by far the most common trap for ML practitioners and provides a number of examples:

1. Deciding on data transformations by looking at the data. Prof. Abu-Mostafa makes the important distinction between ogling the dataset and making use of known properties of the data source. The latter is allowed as it is not subject to the stochastic nature of a particular data set sampled from this source.
2. Normalising data before splitting. This is demonstrated using Puzzle 4 on financial forecasting. Because the data is split **after** normalisation, information from the test set was used in the training set as the normalisation parameters are now also based on the data in the test set.
3. Trying multiple models. Prof. Abu-Mostafa points out that, if you try enough models and you decide on their utility using the same test data, you will eventually find a model that works well. **If you torture the data long enough, it will confess.** Importantly, this may include models that others have tried and that you have discarded on the basis of the unsatisfactory performance reported by others!

Two remedies for data snooping are briefly discussed: you either avoid it, but this is difficult; or you account for it. In the end, the only way to really account for it, is to have a test data set that is only used once to determine the performance of a machine learning model.